

Examiner
only

6. Consumers rely on a continuous supply of electricity from the National Grid system.

(a) Describe how the National Grid provides a reliable supply of electricity as efficiently as possible. [6 QER]

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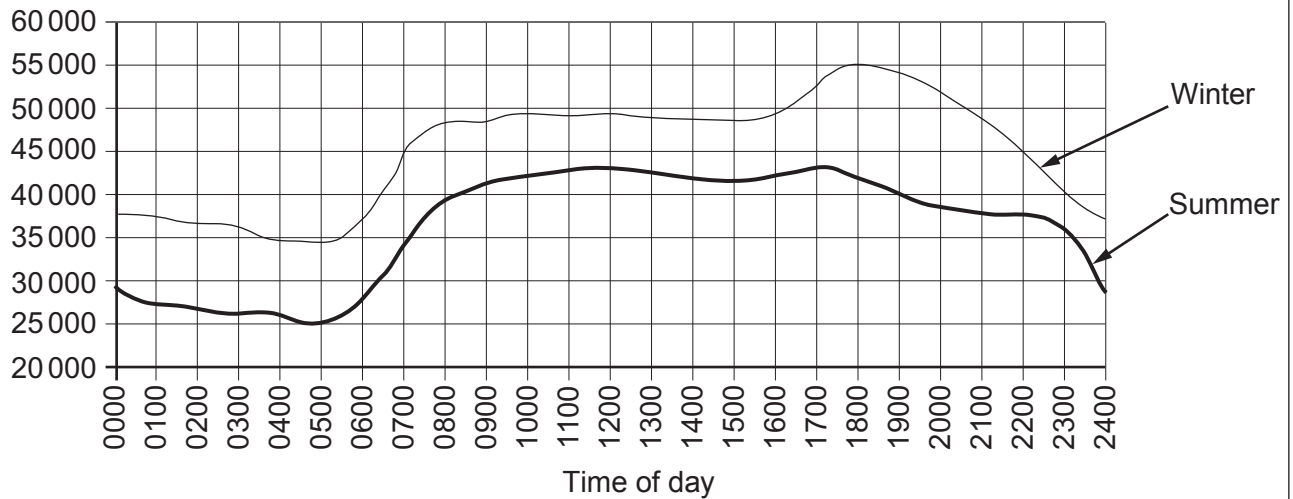
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(b) Information about the National Grid is shown below.

Typical summer and winter demands

National Grid demand (MW)



Network size

- | | |
|--|-----------|
| • Maximum demand | 63 GW |
| • Output capacity | 79.9 GW |
| • Annual mean electrical energy used in the UK | 360 TWh |
| • Length of 400 kV grid | 11 500 km |
| • Length of 275 kV grid | 9 800 km |
| • Length of 132 kV (or lower) grid | 5 250 km |

Energy losses at maximum demand

- | | |
|--|--------|
| • Heating in cables | 860 MW |
| • Fixed losses | 260 MW |
| • Sub-station transformer heating losses | 145 MW |
| • Generator transformer heating losses | 155 MW |

Total losses = 1 420 MW

- (i) Use the information above and an equation from page 2 to calculate the % efficiency of the National Grid system at maximum demand. [3]

% efficiency =



- (ii) Use the equation:

$$\text{power} = \frac{\text{energy transferred}}{\text{time}}$$

to calculate the annual mean power used in GW.
(1 year = 8 766 hours)

[3]

Mean power = GW

- (iii) Tick (✓) the boxes next to the **two** correct statements about the data opposite. [2]

At maximum demand there is spare capacity of 16.9 GW

☐

Maximum demand is equivalent to 85% of total capacity

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The difference in minimum demand between summer and winter is approximately 10 000 MW

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The times of peak demand in summer and winter are **exactly** the same

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The difference in maximum demand between summer and winter is 15 000 MW

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END OF PAPER

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